

Improvisation in Product Innovation: The Contingent Role of Market Information Sources and Memory Types

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Kyriakos Kyriakopoulos

ALBA Graduate Business School, Greece

Abstract

Though improvisation has emerged as a crucial organizational competence, research wrestles with a key question – how can organizations benefit from improvisation? To advance research, I build a richer conceptualization of the role of knowledge resources – market information sources and memory types – in improvisation. I argue that as (a) internal and external sources of market information differ in speed and novelty and (b) procedural and declarative memory differ in articulation and abstractness, they shape the impact of improvisation on new product outcomes in a distinct way. A longitudinal survey of new products in the Dutch food industry uncovers that, contrary to prior thinking, improvisation *increases* cost efficiency when new product teams rely on internal market information and declarative memory and they minimize external market information. Though both sources of market information and low procedural memory *weaken* the negative impact of improvisation on market effectiveness, these conditions were not enough to make its impact positive, echoing traditional concerns with improvisation.

Keywords

improvisation, market information flows, memory, product innovation

Research focuses increasingly on improvisation as a key theoretical lens for studying emergence, innovation and adaptation (e.g. Burgelman & Grove, 2007; Davis, Eisenhardt & Bingham, 2009; Weick, 1998). Building on concepts from diverse fields such as jazz and theatre, conceptual work converges on the extemporaneous nature of improvisation (e.g. Moorman & Miner, 1998a; Vera & Crossan, 2004; Weick, 1998). Empirical studies examine improvisation in various fields such as new product development (Moorman & Miner, 1998b), firm renewal (Brown & Eisenhardt, 1997), technology implementation (Orlikowski, 1996), city administration (Vera & Crossan, 2005), information systems development (Magni, Proserpio, Hoegl & Provena, 2009), or entrepreneurial settings (Baker, Miner & Eesley, 2003).

Corresponding author:

Kyriakos Kyriakopoulos, ALBA, Athinas Ave. & 2A Areos Str., 166 71 Vouliagmeni, Greece.
Email: kkyriako@alba.edu.gr

A key conclusion of this work is that the value of improvisation – its impact on performance outcomes – is not automatic in business settings (Moorman & Miner, 1998a; Vera & Crossan, 2004; Weick, 1998) as implied in earlier research. A core theme in this literature describes how the value of improvisation depends on knowledge resources (e.g. Miner, Bassoff & Moorman 2001; Samra, Lynn & Reilly, 2008). In particular, researchers have shown that information flows and memory (knowledge stocks) shape whether improvisation is valuable or not (Cunha, Cunha & Kamoche, 1999; Moorman & Miner, 1998a; Vera & Crossan, 2005).

Despite these insights, research has yet to develop a more nuanced theory of the role of different knowledge resources in the value of improvisation. I seek to address, in particular, two key theory limitations. Though prior work has explored the moderating role of real-time information *in general* on the premise that it increases the awareness and alertness of the improvising team (Moorman & Miner, 1998a; Vera & Crossan, 2005), research in other fields has shown that the nature and sources of information can variably shape performance (e.g. Ancona & Caldwell, 1992; Dougherty, 1992; Moorman, 1995). This led me to explore how *market* information shapes the value of improvisation. Firms increasingly rely on customer or competition intelligence to guide their decisions (e.g. Dougherty, 1992; Henard & Szymanski, 2001; Moorman, 1995); yet, we lack knowledge of how this market intelligence shapes the value of improvisation.

More importantly, among market information flows, I distinguish *internal sources* (e.g. departments, other teams) from *external sources* (e.g. customers, channel partners). I build on the widely accepted notion that external sources tend to generate novel but difficult-to-use information, while information from internal parties tends to be easier to process but less novel (e.g. Allen, 1970; Ancona & Caldwell, 1992; Dougherty, 1992). This contingent view of different information sources has been asserted in various empirical studies which, however, assume away planning issues (e.g. Moorman, 1995; Tushman & Katz, 1980). Instead, I examine the effect of different market information sources *when* teams improvise. Building on the intuition that the time pressure during improvisation narrows the opportunity for processing information, I explore whether the impact of improvisation varies depending on the source of market information.

Beyond market information, I also explore how *procedural* (routine knowledge) and *declarative* (fact knowledge) memory (Anderson, 1983; Cohen & Bacdayan, 1994) affect the value of improvisation. Though Moorman and Miner (1998a) have theorized the moderating impact of these two memory types on the value of improvisation, their theory needs further development as it examines rather abstract and general firm outcomes of action – speed, novelty and coherence. For example, both memory types are argued to increase action coherence defined as external and internal fit. However, given that internal and external coherence are distinct dimensions driven by different factors (see Henard & Szymanski, 2001), it is crucial to unpack the distinct moderating effect of memory types on the impact of improvisation.

Empirical studies aggravate the lack of theoretical precision as they examine either the moderating role of memory *in general* on the effect of improvisation (Moorman & Miner, 1998b; Vera & Crossan, 2005) *or* the direct effect of procedural and declarative memory, typically overlooking improvisation (Kyriakopoulos & de Ruyter, 2004). I advance the argument that, given their difference in articulation and abstractness, procedural and declarative memory affect the value of improvisation in different ways.

In addition to theory limitations, empirical generalizations are also limited given (a) a lacuna of quantitative studies on improvisation which (b) are biased to study dynamic, high-velocity markets (e.g. Baker et al., 2003; Eisenhardt & Tabrisi, 1995; Orlikowski, 1996) and (c) rely on a small sample of firms (e.g. Moorman & Miner, 1998b; Vera & Crossan 2005). Hence, our capacity to inform practice on the incidence and value of improvisation is crippled. For example, research

implies that improvisation is typical in fast-paced markets while the traditional pre-planned action is crucial in stable, mature markets (Eisenhardt & Tabrizi, 1995; Kamoche & Cunha, 2001). I began this study wondering whether improvisation regularly happens and has value in mature settings. To advance our knowledge, this study was designed to rely on a longitudinal survey of a large sample of food processing firms which forms a rather mature setting.

For theoretical framing, I begin with a short review of improvisation literature to distil its key traits. Next, I introduce the outcomes of improvisation, market information and memory as moderators, and the resulting hypotheses. The study continues with the testing of the hypotheses and concludes with the implications of the key findings.

Improvisation: Definition and Key Issues

The metaphor of jazz and theatre improvisation has provided vivid images and vocabulary for theorizing emergent action in organizations (e.g. Hatch, 1999; Kamoche & Cunha, 2001; Vera & Crossan, 2004; Weick, 1998). This research compares improvisational firm action to improvisational theatre, which neither follows predefined roles nor is a staged event that follows a preconceived plot. Rather, planning emerges through acting on external stimuli, just as jazz musicians adjust to one another's playing and the audience's reactions.

Though metaphors continue to inform research on improvisation (Cornelissen, 2006), organizational research has acknowledged that 'improvisation is more than a metaphor' (Crossan, 1998: 593) and sought to unpack the definition of organizational improvisation and distinguish it from related constructs and phenomena (for extensive reviews see Cunha et al., 1999; Moorman & Miner, 1998a, 1998b). Within this research tradition, I follow Miner et al. (2001) to define organizational improvisation as the merger of composition and execution, real-time action undertaken by teams, units, or entire firms.¹ This definition stresses that when improvising, planning and execution are *not* separated – in contrast to the pre-planning models (Ansoff, 1988) – but they are interwoven both temporally and substantively (Miner et al., 2001). It is also consistent with the work of Vera and Crossan (2005) stressing two key correlates of improvisation, *spontaneity* – extemporaneous or impromptu quality – and *creativity* – departure from existing routines and plans but not necessarily novel outcomes.

A key theme in organizational studies of improvisation (e.g. Vendelø, 2009; Weick, 1998) examines the relationship between improvisation and organizational learning, i.e. the change of organization knowledge in the form of beliefs and/or behaviours (Cyert & March, 1963; Huber, 1991). Research clarifies that they are distinct concepts as learning can be achieved in other ways beyond improvisation such as vicarious learning or trial-and-error efforts (Moorman & Miner, 1998a, 1998b). Accordingly, improvisation is cast as *a* form of online, real-time learning to deal with unexpected events which can help or hurt a firm (Miner et al., 2001). Real-time information flows and organizational memory can trigger or block improvisation as well as shape its impact (Moorman & Miner, 1998a, 1998b; Vera & Crossan, 2005). Research has observed that improvisation affects learning, too, in various ways: it may have ephemeral or even harmful impact unless firms develop improvisational competencies (Cunha et al., 1999; Miner et al., 2001).

Along these broad connections between learning and improvisation, I take interest in how market information flows and stored knowledge affect the value of improvisation (Moorman & Miner, 1998a, 1998b; Vera & Crossan, 2005). Though these studies have generated insights, they need further theoretical precision and empirical investigation as explicated in the introduction. The next section seeks to advance this set of ideas by theorizing the role of a more *refined* typology of knowledge resources in real-time action.

Theory and Hypotheses

To ground theory development, I first present the focal outcomes. Then, I introduce a typology of knowledge resources (market information flows and memory) and the ensuing hypotheses on how knowledge resources shape the value of improvisation.

Focal Innovation Outcomes

New product development (NPD) is a key area in which firms leverage knowledge resources and improvise to deal with unforeseen events (e.g. Kamoche & Cunha, 2001; Magni et al., 2009). Moreover, against the backdrop of limited empirical studies on improvisation, product innovation remains the most prominent research context (Eisenhardt & Tabrizi, 1995; Moorman & Miner, 1998b; Samra et al., 2008). This study, thus, can help us reassess previous findings and provide cumulative evidence, an important priority for the field (Vera & Crossan, 2004).

I explore two crucial outcomes of improvisation in NPD: cost efficiency and market effectiveness. *New product cost efficiency* pertains to production, promotion, and marketing costs as a percentage of sales associated with the new product. Thus, cost efficiency typically relates to process formalization, leveraging prior experience, time efficiency, or local learning (e.g. Olson, Walker & Ruekert, 1995; Wong, 2004). *New product market effectiveness* examines new product sales and market share relative to the firm's major competitor. It has been found to relate to product advantage, innovativeness, technological/marketing expertise and cross-functional and market-oriented learning (Henard & Szymanski, 2001).

These two outcomes represent important performance criteria of product innovation (Henard & Szymanski, 2001). Needless to say, improvisation can have an impact on other outcomes (e.g. creativity or speed); they are, however, intermediate and thus indirectly reflected in cost efficiency and market effectiveness, which are end outcomes. Moreover, our outcomes help us address equivocality in the value of impromptu action. For example, improvisation might appear to undermine efficiency as firms deviate from well-exercised routines and plans (Cooper & Kleinschmidt, 1986) and its attention-consuming nature increases costly errors (Davis et al., 2009). Other research, however, argues that real-time action boosts efficiency as it inherently saves time and resources (Eisenhardt & Tabrizi, 1995; Samra et al., 2008), given its spontaneous nature (Vera & Crossan, 2005).

These conflicting claims indicate the presence of other factors influencing whether improvisation has a positive or a negative effect on NPD outcomes. As the review indicated earlier, I suggest that different types of memory and sources of market information represent a prime candidate for understanding the value of improvisation. I consider this issue next.

Knowledge Resources

Market information flows. Research acknowledges that real-time information increases awareness and alertness during improvisation (e.g. Eisenhardt & Tabrizi, 1995; Vera & Crossan, 2005). Improvising teams are advised to be 'present in the moment' (Hatch, 1999; Spolin, 1963) in order to coordinate their actions (Moorman & Miner, 1998a) and gain immediate feedback on their actions (Brown & Eisenhardt, 1997).

Outside improvisation, there is a long tradition of research on information use in product innovation (e.g. Allen, 1970; Tushman & Katz, 1980; von Hippel, 1988). What stands out in this field is a key distinction between internal (e.g. internal experts, concurrent projects, R&D department, salesforce) and external (e.g. customers, supply chain partners, alliances, outside experts, consultants) sources of information (Huber, 1991). Parallel to the notion of local and distant search

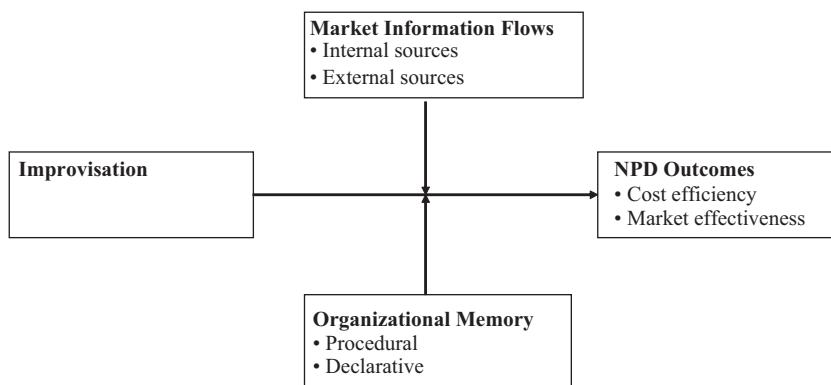


Figure 1. Conceptual Framework.

(Cyert & March, 1963), internal and external sources affect whether communication occurs within established mental models. Specifically, intra-firm communication often rests on shared language, coding schemes, and worldviews that make communication efficient by transmitting large amounts of information with minimal misinterpretations (Katz, 1982; Tushman & Katz, 1980). However, such communication is very likely to involve similar information and interpretation (Wong, 2004).

In contrast, external communication is more likely to draw on new ideas or interpretive frameworks outside the firm's dominant mental model; no wonder, teams are advised to contact directly external parties to break out from entrenched communication filters and worldviews (von Hippel, 1988). External contacts, however, often have an important drawback; it takes time to process such information, given the lack of shared language (Tushman & Katz, 1980).

Improvisation research has referred to similar distinctions in information flows in passing (Cunha et al., 1999); I extend these insights on external and internal sources to theorize how they shape the value of improvisation.² Furthermore, I focus on *market* information that refers to 'data concerned with a firm's current and potential external stakeholders' (Moorman, 1995: 319). Market information plays a critical role in NPD both in marketing and management fields (e.g. Brown & Eisenhardt, 1997; Dougherty, 1992; Henard & Szymanski, 2001; Tushman & Katz, 1980; von Hippel, 1988). For example, information on customer segments can guide product design which, thereby, can shape both cost efficiency and market effectiveness (Moorman, 1995; von Hippel, 1988). Hence, the focus on market information helps us paint a richer portrait than prior research studying the moderating role of information in general.

The moderating effect of internal market information. Market information from internal sources is likely to reflect the mental model of the firm and its key priorities, for example, target segments, product quality and key competitors (Dougherty, 1992; Moorman, 1995). This feature of internal sources will improve the impact of real-time action on cost efficiency for both cognitive and emotional reasons. First, they provide a common context for a team's impromptu actions, which is crucial for developing quickly a shared understanding and aligning team actions around firm priorities (e.g. Wong, 2004). Given the attention-demanding nature of improvisation compared to pre-planned actions, simply the common language and priorities facilitate understanding and reduce the number of possible options. In turn they help teams avoid delays and costly mistakes (Davis et al., 2009), random undirected action (Cunha et al., 1999) or incoherent action (Moorman & Miner, 1998b). This insight resonates with jazz literature, which stresses that pace and coherence

call for communication within the jazz band (e.g. eye contact, hand signals) to 'constrain the turbulence of the jazz process by specifying particular ways of inventing and coordinating musical ideas' (Bastien & Hostager, 1988: 586).

Second, accessing information from other units of the firm helps team members see the commitment of the firm to the ongoing project, which can motivate the team (Eisenhardt & Tabrizi, 1995; Magni et al., 2009). This is crucial to improvisation as it works with limited attention and analysis (Davis et al., 2009), so team members can experience anxiety and uncertainty (e.g. Cunha et al., 1999). Systematic communication with internal parties can increase their confidence that their impromptu actions are grounded in firm strategic priorities. This confidence, in turn, increases the speed of improvisation, as Brown and Eisenhardt (1997) note in their study of the computer industry. In summary, the common context and confidence associated with internally sourced market information increase the speed of action and the product compatibility with internal standards, thereby saving costs.

It is less clear, however, how the same features of internal market information affect the impact of improvisation on market success. Some research acknowledges the beneficial effect of internally generated market insights by facilitating a shared vision of strategy and execution (Jaworski & Kohli, 1993; Moorman, 1995). The intuition is that in order to create superior customer value, a key driver of market success, the NPD team should combine market information from different internal functions, according to total quality management research (Imai, Nonaka & Takeuchi, 1985). This, in turn, can preempt fragmentation or interdepartmental conflict, which are often observed in improvisation (Miner et al., 2001) and hurt market effectiveness. Other research, however, points to the limited diversity of internal sources (e.g. complaints from existing customers, information about current rivals), directing attention to certain areas and ignoring others (Huber, 1991; Katz, 1982). Even if the team is exposed to novel information, entrenched viewpoints filter out novel thinking (Christensen & Bower, 1996).

In evaluating these opposing arguments, I suggest these limitations of internal sources appear typically in studies of pre-planned innovation and dealing with internal information but not internal *market* information (e.g. Wong, 2004). In pre-planned innovation, the temporal separation of planning and execution can render information from internal sources obsolete as markets evolve. During improvisation, however, *market* information from internal parties can integrate the team actions around key market priorities. This, in turn, increases the odds of creating superior customer value, a key driver of market success (Olson et al., 1995). Hence:

Hypothesis 1: Internal market information moderates *positively* the effect of improvisation on (a) cost efficiency and (b) market effectiveness.

The moderating effect of external market information. Market information from external sources tends to exhibit diversity and offer fresh perspectives (Allen, 1970; Moorman, 1995). This feature of external sources creates a set of different effects on the value of improvisation. I begin with the prediction that external sources will hurt the impact of improvisation on cost efficiency because of interpretation and integration difficulties (Dougherty, 1992; Tushman & Katz, 1980). External sources likely trigger varied understanding and hence team members must discuss longer to reach consensus (Wong, 2004). This is due to differences in the coding schemes and technical language between project members and external parties, making the interpretation of external information time consuming (Tushman & Katz, 1980).

Moreover, external communication can result in improvisations that are inconsistent with the firm's manufacturing systems or marketing priorities, which raises production and marketing costs.

Innovation research observes the difficulties of integrating externally sourced information with ongoing firm activities (e.g. Lee & Allen, 1982) and reveals a negative relation between external communication and project efficiency, such as coherence, adherence to budget, or meeting a schedule (e.g. Ancona & Caldwell, 1992). In pre-planning, teams have time to adapt the external information to their internal systems; the time pressure of extemporaneous action, however, limits thoughtful analysis and action on this new information. Baker et al. (2003), for example, describe the constraints on the operations of knowledge-based start-ups because of improvisations aimed to legitimate their actions to outside resource providers.

Yet, external sources can enhance the impact of improvisation on market success because the improvising team members have the opportunity to see the change emerge in its context without delays or internal filters. Specifically, feedback from external parties can help the team realistically evaluate alternative designs and adjust product definition to evolving market conditions (Brown & Eisenhardt, 1997; Moorman & Miner, 1998b). In this process, members confront different world-views and thus become conscious of their tacitly held assumptions, challenge entrenched recipes, and develop original associations (Wong, 2004).

This external feedback also carries strong emotional benefits as it reduces the risky nature of innovation, which is even more pronounced during improvisation; team members feel more confident in their choices as they are grounded in market reactions and not driven solely by abstractions. This proactive approach to the market also fosters a risk-taking attitude and experimentation (Eisenhardt & Tabrizi, 1995). Little wonder, then, that the innovation literature suggests an innovation team needs direct contact with lead customers to break away from their established thinking (Mintzberg, Pascale, Goold & Rumelt, 1996; von Hippel, 1988).

Still, other research is more cautious on the benefits of external sources as firm beliefs and mental maps tend to persist, i.e. cognitive inertia, across firms in the same industry even in the presence of disconfirming information (Daniels, Johnson & de Chernatony, 2002). Christensen and Bower (1996) recount how firms in the disk drive industry with close relations to powerful customers became oblivious to disruptive technological change in favour of more incremental innovations. Despite these different views, in general, direct contact with external parties is reported to promote novel thinking and market success (e.g. Henard & Szumanski, 2001; Moorman, 1995; von Hippel, 1988). Moreover, these cautionary notes are more applicable in settings of radical and disruptive change which are not the focus of the present study. Thus:

Hypothesis 2: External market information moderates (a) negatively the effect of improvisation on cost efficiency and (b) positively the effect of improvisation on market effectiveness.

Memory. In addition to market information flows, organizational memory in the form of skills, routines, or stocks of knowledge³ are also crucial for valuable improvisation (Kamoche & Cunha, 2001; Spolin, 1963; Weick, 1993). Prior empirical work considers the role of memory in general (Moorman & Miner, 1998b; Vera & Crossan, 2005) while conceptual work hypothesizes the contingent role of procedural and declarative memory in the value of improvisation for general firm outcomes – action novelty, speed and coherency (Moorman & Miner, 1998a). I seek to advance this research by exploring the moderating role of procedural and declarative memory in the impact of improvisation on cost efficiency and market effectiveness.

Procedural knowledge refers to memory about how things are done (Anderson, 1983; Cohen & Bacdayan, 1994) or ‘things you can do’ (Berliner, 1994: 102) and thus may be referred to as motor memory (Anderson, 1983). Procedural memory at the organizational level is embodied in routines involving ‘patterned sequences of learned behaviour involving multiple actors who are linked by

relations of communication and/or authority' (Cohen & Bacdayan, 1994: 555). Because of its routine nature, procedural memory takes the form of skills that are tied to the particular domain in which they are exercised (Moorman & Miner, 1998a).

In NPD, it includes routines for team cooperation, project milestones (Moorman & Miner, 1997) or skills in processes such as screening, development and launch (Samra et al., 2008). Moreover, because of its automatic and tacit nature, procedural memory is often available to users, even if they lack the underlying substantive knowledge; a phone number can be hard to be stated but becomes available when the phone is at hand (Cohen & Bacdayan, 1994).

Declarative memory refers to memory for facts, events, or propositions (Cohen & Bacdayan, 1994) and it can be consciously or intentionally recollected (Anderson, 1983; Moorman & Miner, 1998b). For example, the procedural memory of riding a bike can be contrasted to the declarative memory of the mechanical principles underlying riding a bike. Unlike procedural memory, declarative memory is more general (not committed to a specific use); so in the NPD context, it might include knowledge about customer preferences, technical expertise, or know-why (Kogut & Zander, 1992), i.e. cause-and-effect relationship between a pricing scheme and market response. Because of the general nature of declarative memory, it can have many applications. However, this abstract nature of declarative knowledge tends to slow down action (Cohen & Bacdayan, 1994).

Building on the automatic nature of procedural memory and the abstract nature of declarative, we now turn to the hypotheses that offer a nuanced portrait of the effects of memory on the value of improvisation.

The moderating effect of procedural memory. Routines are likely to boost the cost efficiency of improvisation when team members apply existing procedures in a new context or recombine existing routines. Miner et al. (2001) describe how a NPD team at a food company similarly drew on its standard process, which involved visiting stores to sell products similar to its own, to improvise a process for selling *dissimilar* products. Outside the product innovation literature, Hutchins (1991) describes how a ship's crew quickly improvised a standard computation sequence procedure in a different context, rescuing a ship with a broken navigational system.

In these and other examples, preexisting procedures act as referents (Miner et al., 2001) or ready-mades (Vera & Crossan, 2005) which provide a pattern of learned (Cohen & Bacdayan, 1994) and legitimate action (Cyert & March, 1963; Walsh & Ungson, 1991) to compensate for the lack of structure inherent to improvisation. Without such memory, member roles, communication and tools for action would need to be built from scratch, as research outside product innovation has shown. Weick (1993), for example, recounts the collapse of routines in the Mann Gulch fire disaster, which upset the role structure of the firefighting team, curbed the legitimacy of orders, and threw the team into disarray. Jazz improvisation research also observes that a standard vocabulary, melodies, or chords provide structure which limits the likelihood of uncoordinated effort or chaos in jazz bands (Berliner, 1994; Kamoche & Cunha, 2001). Building on these insights, empirical research in NPD has shown that a combination of a structured approach and improvisation increases cost and time efficiency (Moorman & Miner, 1998a; Samra et al., 2008).

At the same time, the unconscious quality of routines increases the likelihood that improvisation will hurt market performance. Because routines are automatic or habitual, procedural memory can be detrimental for the value of improvisation in shifting market demands. As team members rely on tacitly held knowledge to create their real-time response, they become oblivious to nonstandard customer needs or competitive reactions (Dougherty, 1992). Cohen and Bacdayan (1994) illustrate

this risk when they remind us that vigilance loss is not unusual for pilots, who are so accustomed to answering each safety check with 'okay,' they fail to see a problem when it actually exists.

In pre-planned innovation, the team has the time to revisit its prior routines and creatively recombine them to match the new situation. In improvisation, the limited response time often leads to blind adherence to routines that may no longer match the new context (Moorman & Miner, 1998a). This idea echoes warnings from jazz about the 'habituated and uninspired use' of standard 'vocabulary patterns' (Berliner, 1994: 134). Thus:

Hypothesis 3: Procedural memory moderates (a) *positively* the effect of improvisation on cost efficiency and (b) *negatively* its effect on market effectiveness.

The moderating effect of declarative memory. Research is not clear on how the abstract and deliberate nature of declarative knowledge will affect the impact of improvisation on cost efficiency. To begin with, declarative memory is not easily actionable because it tends to be abstract (Anderson, 1983; Cohen & Bacdayan, 1994). During improvisation, declarative knowledge slows the integration and conversion of expertise into product specifications. This can increase production costs because the product design ultimately might not support cost-efficient production methods. Researchers tend to consider this difficulty part of the split between design engineers and production engineers, such that production engineers believe design engineers with strong theoretical knowledge create problems by drawing on abstract knowledge instead of relying on readily available heuristics (Dean & Susman, 1989).

On the other hand, declarative memory can improve the likelihood that improvisation boosts cost efficiency as firms do not need to 'reinvent the wheel' and they easily leverage old knowledge to improvise in new settings. Declarative knowledge also helps teams improvise actions that are coherent with the firm's expertise and capabilities (Moorman & Miner, 1998a). Leveraging production knowledge, for example, could significantly reduce the manufacturing costs of new products as the design of the new product fits production requirements. Consistent with this, new product development studies concur that marketing and technological expertise improve efficiency in new products (e.g. Cooper & Kleinschmidt, 1986; Henard & Szymanski, 2001). Following this theoretical work and empirical evidence, I suggest that declarative memory will strengthen the impact of improvisation on cost efficiency.

Turning to the impact of improvisation on market effectiveness, the abstract nature of declarative memory is likely to play a positive role because it helps teams identify patterns in external events and select actions suited to those conditions (Moorman & Miner, 1998a). Moreover, because declarative knowledge applies in a variety of contexts (Cohen & Bacdayan, 1994; Kogut & Zander, 1992), it can provide a rich pool of options to help the team improvise (Vera & Crossan, 2005). Outside business settings, Weick (1993) describes how the leader of a firefighting team could rely on extensive knowledge about fire emergencies to improvise an escape fire to open a safe path to escape. Effective jazz performances similarly rest on knowledge of jazz music theory, history, instrumentation and music technology – all forms of declarative knowledge (Berliner, 1994). In organizational settings, NPD studies suggest that marketing and technological expertise improve new product success (e.g. Henard & Szymanski, 2001). This benefit of declarative knowledge is important for many forms of action, but it becomes crucial during improvisation because of the limited time the team has to search for knowledge before it takes action. Hence:

Hypothesis 4: Declarative memory moderates *positively* (a) the impact of improvisation on cost efficiency and (b) the impact of improvisation on market effectiveness.

Methods and Measures

Research Setting

The choice of research setting was guided by two criteria. First, prior research tends to study improvisation in turbulent settings, for example, navigation system failure (Hutchins, 1991), entrepreneurial settings (Baker et al., 2003) or high-tech industries (e.g. Eisenhardt & Tabrizi, 1995; Magni et al., 2009; Samra et al., 2008). These settings share extremely high levels of uncertainty and turbulence, making extemporaneous action a frequent and valuable practice. I chose to research improvisation in the food industry as I wondered whether improvisation regularly happens and has value in a rather mature, predictable setting. Second, systematic quantitative studies are rare and even these rely on data from a *few* organizations. For example, Magni et al. (2009) collected data from 38 information systems development teams from a single firm; Moorman and Miner (1998) collected data on NPD actions from two firms; Vera and Crossan (2005) collected data from 50 work teams in a single municipal setting. There is not, hence, cumulative evidence and prior findings are not easily generalized in other contexts.

Accordingly, a sample of 500 Dutch business units from the food processing industry was randomly selected from a complete business directory of Dutch companies. A total of 160 firms noted they did not perform NPD activities (active in wholesaling or retailing), so the actual sample consisted of 340 firms. Of this eligible sample, 136 returned questionnaires, of which 4 were incomplete. Thus, the usable sample is 132 firms for a response rate of 39%. I employed marketing managers as key informants because of their deep involvement with product development and market information (e.g. Jaworski & Kohli, 1993; Olson et al., 1995), given the critical role of marketing in NPD in the food sector (Traill & Meulenbergh, 2002). When these managers were not involved in product development, they were asked to pass on the survey to knowledgeable project managers. In the end, 64% of respondents are project managers identified by the marketing managers of the initial mailing. Respondents report high levels of involvement with the project ($M = 5.97$, $SD = 1.65$, 7-point scale). Finally, there are no systematic differences between respondents before and after the second mailing.⁴

The questionnaire asked informants to focus on a NPD project (our unit of analysis) completed in the past 12 months. When asked to identify the newness of the project, 17% of the respondents described it as a completely new product with a new market, 24.5% as a new product line, 38.3% as an addition to an existing line, 9.6% as a repositioning of an existing product, and 10.6% as an improvement of an existing product. In an effort to establish causality among the variables, one year later, I mailed these 132 respondents another questionnaire focusing on the market performance of the new product. This mailing generated 75 responses, but after discarding 15 incomplete questionnaires, there are 60 usable cases for a 45% response rate. It is worth noting that 55% of these respondents are different from the initial respondents (who had left the company or moved to another unit or had no time to answer). According to best practices (Podsakoff, MacKenzie, Lee & Podsakoff, 2003), the time lag and different respondents strengthen the quality of our inferences for containing common method variance (CMV). Finally, I find no systematic differences between firms that did and did not respond after a year.⁵

Measures

Explanatory variables. I employed existing scales to tap the improvisation, memory and information flow constructs (see the appendix for the measures verbatim). I used the scale of Moorman and Miner (1998b) to measure team-level *improvisation*, which captures the degree of simultaneity of

planning and execution without being confounded with correlates of improvisation such as innovation or bricolage. To tap the *external and internal market information flows*, I employed scales developed by Kyriakopoulos and de Ruyter (2004). To ensure respondents consider internal information flows as distinct from firm memory, they were asked to focus on information about ongoing market developments they received during the project from various internal sources.

After looking to the literature for measures of memory types, it became clear there is not a stark, absolute distinction between procedural and declarative memory; for example, prototyping skills contain both procedural and theoretical technical knowledge underlying these skills. To alleviate this concern I devised measures that tend to exhibit a dominant emphasis after feedback first from academic experts and then from marketing managers. In the example of prototyping skills, procedural skills dominate technical knowledge. This principle was applied to items from Moorman and Miner (1997, 1998b; see also Kyriakopoulos & de Ruyter, 2004) presented to a panel of respondents during pre-testing.

Following this process, because *procedural memory* centres on well-exercised procedures and processes, in the form of skills, I asked the informants to rate the extent to which 'For this project, my team relied on: well-defined procedures/a standard approach/strong skills.' *Declarative memory* pertains to facts, events and propositions, so informants indicated the extent to which 'For this project, my team relied on: a great deal of knowledge/strong expertise/knowledgeable people.' For example, according to respondents' feedback during pretesting, the term 'expertise' describes a good understanding of or ability to analyse specific areas, such as the environment, technology, or customers. This is consistent with our definition.

New product outcomes. The measure of *new product cost efficiency* focuses on the production, marketing and promotion costs of the new product as a percentage of sales and relative to industry averages. These benchmarks ease biases related to the variety of project objectives.

I measured market performance at two distinct points of time. First, with *new product market effectiveness after one year*, I examine new product sales and market share outcomes relative to a firm's major competitor one year after the product launch. Given the cross-sectional nature of that measure and as a way to elucidate the causal relationships among the key variables, one year later, I assess *new product market effectiveness after two years* involving these same outcomes two years after launch. Measuring performance as the success of a firm relative to its most competitive rival eliminates the distortion created by product/market differences.

In addition, I tried to collect direct quantitative measures such as volume or value of sales, but respondents were not eager to report such confidential information. It is worth noting, finally, that the correlation (Table 1) between the two measures of market effectiveness is high ($r = .64, p < .01$) indicating that the market success of new products changed little within a year.

Control variables. I control for the size of a business unit's *firm resource level* as larger firm size might explain the variance in product outcomes. To ensure that the results are not a function of the change in the firm's environment, *competitive intensity* is controlled for. Finally, *project newness* is controlled for as it may affect the NPD outcomes (see appendix).

Common method variance test. Beside the procedural steps to contain CMV, two sets of tests were applied. First, I relied on Harman's one-factor test to determine if common method bias exists (Podsakoff et al., 2003). The result of the principal components factor analysis revealed 10 factors with eigenvalues greater than 1.0, which corresponds to the 10 variables in our model. Because the results indicate that no general factor emerged in the unrotated factor structure, common method

bias does not appear to be a concern. I performed the same test on pairs of independent and dependent variables and found it consistently produced two distinct factors.

Second, I include a marker variable (respondent involvement with the project) which is theoretically unrelated to the focal constructs (Lindell & Whitney, 2001). All unadjusted correlation coefficients remain significant at $p < .05$ after the marker variable was partialled out. Though both tests do not rule out CMV, they indicate that the findings are less likely due to method bias.

Measure Purification

Following the data collection, the scales were purified according to existing recommendations (e.g. Bagozzi & Yi, 1988). On subsets of theoretically related variables (Bagozzi & Yi, 1988), I ran two confirmatory factor models that feature the focal dependent variables on the one hand, and new product outcomes and control variables on the other hand. Any items with large modification indices or weak loadings or that load on more than one factor were deleted (see the appendix), if doing so did not reduce the domain coverage. Overall, the results demonstrate adequate levels of fit for both the first model ($\chi^2 = 42.19$, NNFI = .90, CFI = .92, SRMR = .063) and the second model ($\chi^2 = 37.50$, NNFI = .89, CFI = .91, SRMR = .075). After dropping the items with low loadings, all factor loadings are well above .40 and significant ($p < .01$). In addition, the average variance extracted by each measure exceeds the recommended cut-offs (Bagozzi & Yi, 1988), which, together with the adequate (Table 1) reliabilities (.67 to .95), establishes the internal consistency of the measures.

I also checked the external validity of the memory measures by examining their correlations with other variables that are distinct from but theoretically related to these memory types. Specifically, I employed the organizational culture scale from Moorman (1995), which taps firm culture, whether it is formalized or organic. Mechanistic structures tend to be associated with routines and formal procedures (Moorman, 1995). Therefore, it is reasonable to expect that procedural memory is more likely to correlate with formalized cultures than with organic ones. Indeed, procedural memory correlates positively with formalized ($b = .34$, $p < .05$) but not with organic cultures

Table 1. Measure Characteristics and Correlation Matrix

	Mean	SD	1	2	3	4	5	6	7	8	9	10
1 Improvisation	3.99	1.45	.84									
2 Internal market information flows	4.50	1.20	-.03	.76								
3 External market information flows	4.45	1.26	.04	.19*	.75							
4 Procedural memory	3.88	1.61	-.24**	.12	.23*	.75						
5 Declarative memory	5.07	1.32	-.22*	.15†	.30**	.16†	.86					
6 Firm resources	4.44	1.23	-.12	.11	.04	.06	.25**	.70				
7 Competitive intensity	4.51	1.34	.06	.02	.03	-.01	.01	.05	.76			
8 Cost efficiency	3.08	1.15	.13	.00	-.07	-.06	-.03	-.05	-.19*	.67		
9 Market effectiveness after one year	4.62	1.62	-.17†	.23**	.28**	.11	.39**	.24**	.10	.01	.95	
10 Market effectiveness after two years	4.75	1.54	-.20+	.15	.35**	.12	.35**	.19†	.09	.02	.64**	.85

** $p < .01$

* $p < .05$

† $p < .10$

Note: reliabilities appear on diagonal in *italics*

($b = -.12$, NS). Without any prior theory claims, declarative memory correlates positively with organic ($b = .24$, $p < .05$) but not with formalized cultures ($b = -.09$, $p > .10$).

To assess discriminant validity, two tests were used (Bagozzi & Yi, 1988). Within each subset, I assessed pairs of constructs in a series of two-factor confirmatory factor models. Specifically, each model was run twice, first constraining the correlation between the two latent variables to unity and then freeing this parameter. The results provide evidence of discriminant validity ($\Delta\chi^2_{(1)} > 3.84$), as does the statistical difference of all ϕ from 1. In addition, in examining the two memory types, two sources of information flow, and two new product outcomes, the average variance extracted for each construct (.47 to .73) is greater than the squared correlation (.003 to .14) between that construct and any other.

Tests of Hypothesized Relationships

To explore the hypotheses, two tests were used. First, I ran three hierarchical regression models with new product cost efficiency and new product market effectiveness after one and two years as the dependent variables. The controls, direct effects of explanatory variables, and two-way interactions entered the models incrementally to indicate their relative contributions. The largest variance inflation factor, an indicator of multicollinearity, is 1.8, far below the 10.0 cut-off, which shows that multicollinearity is not a major concern. Out of caution, following Cohen, Cohen, West, and Aiken (2003), I nonetheless mean-centred the variables before forming the interactions to reduce any potential problems linked with multicollinearity.

Next, I probed the significant two-way interactions by conducting simple slope tests for low (one standard deviation below the mean) and high (one standard deviation above the mean) levels of the moderating variables (Cohen et al., 2003). I then plotted the slopes from the probing in figures to facilitate interpretation. The purpose of this analysis is to examine whether the moderators make improvisation valuable or not for each outcome.

Results

Overall Model Results

As shown in Table 2, model 3 with the two-way interactions is significant as it pertains to new product cost efficiency (adjusted $R^2 = .08$, Change-in- $F_{(12,119)} = 3.117$, $p < .01$), market performance after one year (adjusted $R^2 = .23$, change-in- $F_{(12,109)} = 2.17$, $p < .10$), and market performance after two years (adjusted $R^2 = .22$, change-in- $F_{(9,51)} = 2.658$, $p < .05$). It is worth noting that the direct effect of improvisation is not significant for cost efficiency ($b = .11$, NS) nor for market effectiveness ($b = -.16$, NS; $b = -.15$, NS).

The Moderating Effect of Internal Market Information Flows (H_1)

H_{1a} theorized that internal market information would enhance the impact of improvisation on new product cost efficiency. The results fail to support this claim ($b = .11$, NS).

H_{1b} predicted that internal sources would have a positive moderating effect on the impact of improvisation on market performance after one year which is supported by the results ($b = .16$, $p < .05$). In probing this interaction, the relationship between improvisation and market performance is significantly negative ($b = -.34$, $p < .05$) for low internal market information but becomes insignificant ($b = .03$, NS) for high levels. The same pattern appears after repeating the analysis for market performance after two years ($b = .37$, $p < .01$). In probing this interaction, the relationship

Table 2. The Effect of Improvisation, Market Information Flows and Memory on New Product Cost Efficiency and Market Effectiveness

Independent variables	Cost efficiency			Market effectiveness after 1 year			Market effectiveness after 2 years†		
	Model 1 B (SE)	Model 2 B (SE)	Model 3 B (SE)	Model 1 B (SE)	Model 2 B (SE)	Model 3 B (SE)	Model 1 B (SE)	Model 2 B (SE)	Model 3 B (SE)
Firm resources	-.07 (.10)	-.08 (.09)	-.07 (.09)	.42 (.15)**	.15 (.15)	.15 (.10)	.24 (.15)	.07 (.17)	.16 (.17)
Competitive intensity	-.18 (.10)†	-.19 (.10)†	-.16 (.09)†	.20 (.11)†	.08 (.11)	.10 (.10)	-.28 (.16)	-.18 (.17)	-.04 (.17)
Project newness	-.09 (.10)	-.11 (.10)	-.11 (.10)	-.14 (.13)	-.07 (.12)	-.10 (.12)	-.03 (.14)	.04 (.15)	-.06 (.15)
IMPR		.15 (.08)†	.11 (.08)		-.11 (.10)	-.16 (.10)		-.17 (.14)	-.15 (.13)
IMIF		.03 (.10)	.03 (.10)		.24 (.12)*	.28 (.12)**		.10 (.18)	.10 (.18)
EMIF		-.10 (.10)	-.17 (.10)†		.24 (.12)*	.18 (.13)		.32 (.16)*	.41 (.16)**
PM		-.03 (.08)	-.03 (.08)		-.04 (.09)	-.03 (.09)		-.14 (.12)	-.20 (.12)†
DM		.01 (.10)	.09 (.10)		.29 (.11)*	.28 (.11)**		.07 (.15)	-.02 (.12)
IMPR * IMIF			.11 (.07) H_1 (+)			.16 (.08)* H_1 (+)			.37 (.13)**
IMPR * EMIF			-.19 (.06)** H_2 (-)			-.02 (.08) H_{3b} (+)			.15 (.07)*
IMPR * PM			.07 (.05) H_3 (+)			-.13 (.06)* H_{3b} (-)			-.13 (.07)
IMPR * DM			.12 (.06)* H_{4a} (+)			.06 (.08) H_{4b} (+)			-.12 (.10)
Adjusted R ²	.01	.01	.08	.07	.18	.23	.05	.09	.19
Change-in-F	1.519, NS	.87, NS	3.117**	3.65*	4.65**	2.173†	1.86, NS	1.62, NS	2.66*
N	132	132	132	122	122	122	60	60	60

** $p < .01$

* $p < .05$

† $p < .10$

‡ The hypotheses for market effectiveness after 2 years are the same as the hypotheses for market effectiveness after 1 year

Notes: † report the unstandardized coefficients as I use mean centring. Standard errors are reported in parentheses

DM: declarative memory, EMIF: external market information flows, IMIF: internal market information flows, IMPR: improvisation, PM: procedural memory

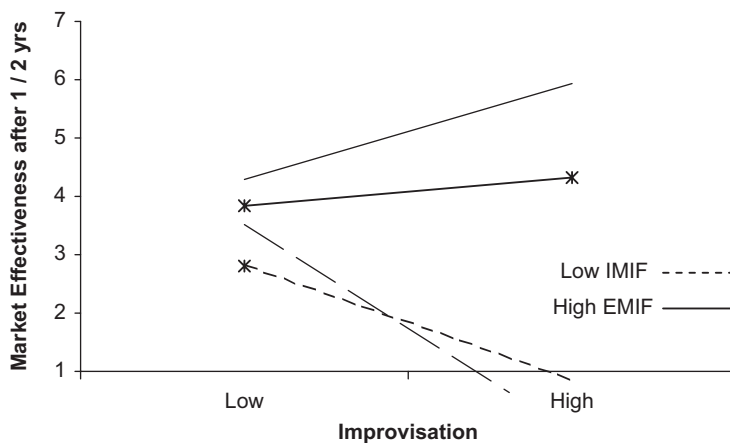


Figure 2. The Moderating Effect of Internal Market Information Flows on Market Effectiveness after 1 and 2 (denoted with *) Years (H_{1b}).

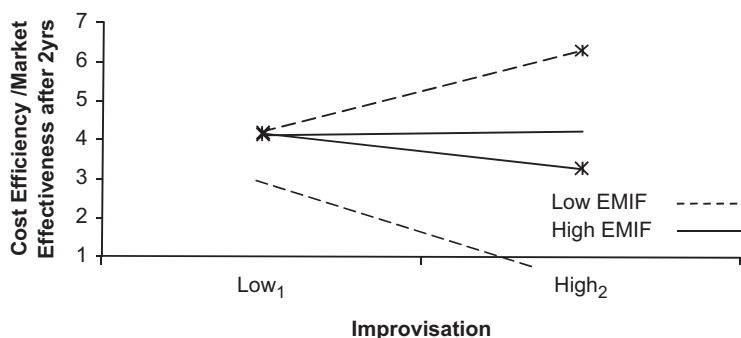


Figure 3. The Moderating Effect of External Market Information Flows on Cost Efficiency (H_{2a}) and Market Effectiveness (denoted by *) after 2 Years (H_{2b}).

between improvisation and market performance is also negative ($b = -.59, p < .01$) at low levels of internal market information but insignificant ($b = .29, NS$) at high levels. As shown in Figure 2, the positive moderating role of internal sources *removes* the negative effect of improvisation on market effectiveness but it was not strong enough to make this effect positive.

The Moderating Effect of External Market Information (H_2)

In considering the influence of external market information, H_{2a} theorized that it would weaken the effect of improvisation on cost efficiency; the results support this prediction ($b = -.19, p < .01$). Through probing, the relationship between improvisation and cost efficiency is significantly positive ($b = .36, p < .01$) at low levels of external market information but insignificant ($b = -.09, NS$) at high levels. Hence, improvisation has a positive effect on cost efficiency in conditions of *low* external market information (see Figure 3).

H_{2b} predicted that external market information would strengthen the effect of improvisation on market effectiveness; the results fail to support this claim for market effectiveness after one year

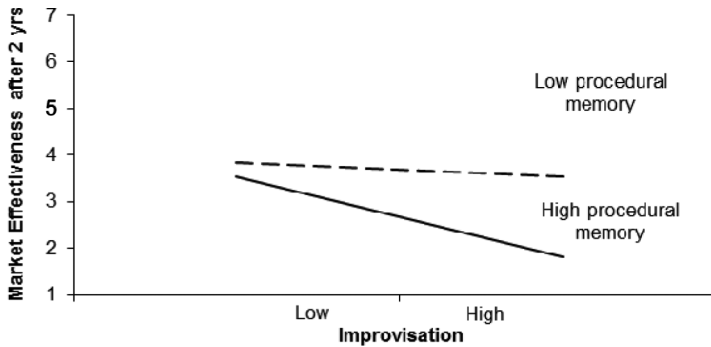


Figure 4. The Moderating Effect of Procedural Memory on Market Effectiveness after 1 Year (H_{3b}).

($b = -.02$, NS), but they support it for market effectiveness after two years ($b = .15$, $p < .05$). In probing this significant interaction, I find that the relationship between improvisation and market effectiveness after two years is significantly negative ($b = -.34$, $p < .05$) at low levels of external market information but insignificant ($b = .03$, NS) at high levels (Figure 3). Internal market information, hence, *removes* the negative effect of improvisation on market effectiveness but does not make this effect positive.

The Moderating Effect of Procedural Memory (H_3)

I theorized that procedural memory strengthens the effect of improvisation on cost efficiency, but the results do not offer support for H_{3a} ($b = .07$, NS). In contrast, procedural memory weakens the effect of improvisation on market performance, in support of H_{3b} ($b = -.13$, $p < .05$). According to the follow-up analysis, the relationship between improvisation and market performance is insignificant ($b = .05$, NS) for low procedural memory and significantly negative ($b = -.37$, $p < .01$) for high procedural memory (see Figure 4). Therefore, improvisation has a *negative* effect on market effectiveness in strong procedural memory. The results associated with market performance after two years fail to support the hypothesis ($b = -.13$, NS).

The Moderating Effect of Declarative Memory (H_4)

Finally, H_{4a} theorized that declarative memory strengthens the effect of improvisation on cost efficiency; the results offer support to the hypothesis ($b = .12$, $p < .05$). The analysis shows that this effect is insignificant ($b = -.04$, NS) for low levels of declarative memory but significant and positive ($b = .26$, $p < .05$) for high levels (Figure 5). Thus, in the presence of strong declarative memory, improvisation has a *positive* effect on cost efficiency; this effect disappears for low levels of declarative memory. In contrast with H_{4b} , declarative memory does not affect the impact of improvisation on market performance for either measure ($b = .06$, NS; $b = -.12$, NS).

Robustness Checks

Two tests were performed to assess the robustness of our results. First, there are fewer usable observations for the model of market effectiveness after two years compared to the model of cost efficiency and market effectiveness after one year. Hence, I ran these latter models on the 60

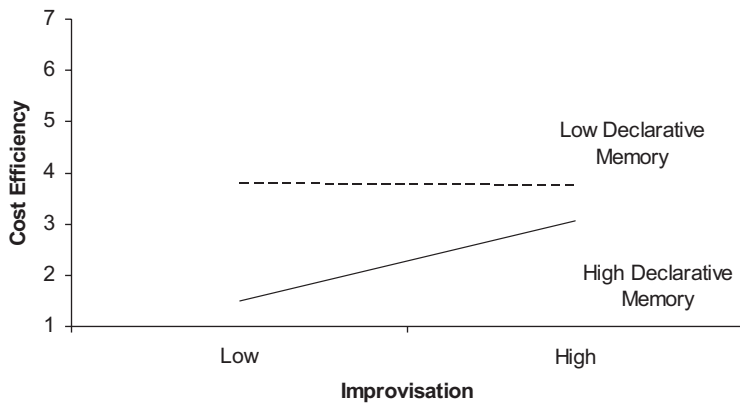


Figure 5. The Moderating Effect of Declarative Memory on Cost Efficiency (H_{4a}).

observations for market effectiveness after two years. For the cost efficiency model (model 3): external market information $\times$ improvisation ($b = -.22$, $SE = .08$) and declarative memory $\times$ improvisation ($b = .13$, $SE = .06$). For market effectiveness after one year (model 3): internal market information $\times$ improvisation ($b = .21$, $SE = .09$) and procedural memory $\times$ improvisation ($b = -.14$, $SE = .06$). Hence, the earlier reported results are robust in this sample.

Second, I explored the generalization of the results to some widely acknowledged boundary conditions of improvisation. Among others, environmental turbulence and project traits might affect the strength of the interactions, so that the effect of memory on the improvisation–cost efficiency relationship could be less beneficial at high levels of turbulence because of the rapid depreciation of prior knowledge in such markets. To explore this possibility, I employed a split group analysis and ran the same regression models at high and low levels of two boundary conditions (Cohen et al., 2003): competitive intensity and project newness. None of the interactions (see Table 3) show any material difference from the general sample results except for the marginally significant interaction of EMIF $\times$ improvisation for market effectiveness after one year ($t = 1.685$, $p < .10$) between novel projects (completely new product, new market, new product lines) versus less novel projects (addition to existing line, improvement of existing product, repositioning of existing product, existing product at lower cost). This analysis, overall, supports the persistence of the findings across different boundary conditions.

Discussion

Summary of Results and Core Contribution

How can firms benefit from improvisation? I find that (a) external market information weakens the impact of improvisation on cost efficiency, while both sources strengthen its impact on market effectiveness and (b) declarative memory increases the impact of improvisation on cost efficiency while procedural memory lowers its impact on market effectiveness. This study, overall, advances the contingent view of the value of improvisation (e.g. Miner et al., 2001; Vera & Crossan, 2005; Weick, 1998) by showing improvisation has conditional value depending on a *finer typology* of knowledge resources. I highlight here three core contributions.

First, I qualify prior arguments that real-time information has a positive role on the premise it increases the awareness and the alertness of the improvising team. My emphasis on *market*

Table 3. Robustness Analysis of the Hypothesized Interactions of Improvisation and Knowledge Resources

Boundary conditions	Interactions	Dependent variables								
		Cost efficiency			Market effectiveness after 1 year			Market effectiveness after 2 years		
		High ^a	Low	t-values	High	Low	t-values	High	Low	t-values
Project newness	IMPR × IMIF	.194	.022	1.031	.301	.088	1.071	.056	.256	.381
	IMPR × EMIF	-.161	-.189	.156	.181	-.178	1.685*	.144	-.107	.593
	IMPR × PM	.030	.109	.0553	-.092	-.021	.042	-.380	.045	1.178
	IMPR × DM	.110	.119	.0034	-.388	.079	1.648	-.037	-.198	.313
Competitive intensity	IMPR × IMIF	.113	.136	.135	.239	.074	.859	.511	.237	.753
	IMPR × EMIF	-.189	-.173	.106	.096	-.136	1.479	.356	.037	1.392
	IMPR × PM	.083	.013	.563	-.223	-.021	1.445	-.074	-.098	.099
	IMPR × DM	.079	.116	.260	-.025	.045	.426	-.065	-.130	.301

* $p < .10$

^a Numbers in the high and low columns are *unstandardized* b-coefficients of the interactions of improvisation with knowledge resources for each new product outcome under high and low level of boundary conditions
DM: declarative memory, EMIF: external market information flows, IMIF: internal market information flows, IMPR: improvisation, PM: procedural memory

information and its different sources uncovered its multifaceted role during improvisation. I have offered new explanations to account for this role. The unifying context and commitment associated with internal sources increase the focus and the coherence of impromptu actions and thereby foster both NPD outcomes. Because of their diversity, external sources increase the likelihood of incoherent action and delays to curb cost efficiency but at the same time they increase the novelty of improvisation boosting market effectiveness.

Second, while Moorman and Miner (1998a) have theorized contrasting effects of procedural and declarative memory on the speed, coherence and novelty of improvisation, I unpack the distinct moderating effect of procedural and declarative memory on different innovation outcomes of improvisation. I have argued that procedural memory provides structure to increase the internal coherence and speed of impromptu actions and thereby cost efficiency, but its automatic nature hurts the external coherence of impromptu actions, thereby undermining market effectiveness.

Finally, this study in the food industry shows that improvisation is not a phenomenon restricted to fast-paced settings, as prior studies tended to assume (e.g. Eisenhardt & Tabrizi, 1995). In fact, firms in mature industries engage in improvisation regularly during NPD and they can benefit from it when they manage skilfully the various knowledge resources. The increasing variety in consumer preferences and competitive action in the food industry create windows of market opportunity (Traill & Meulenberg, 2002) which can be harvested with careful management of improvisation. The appreciation of the benefits and liabilities of impromptu action in mature industries such as the food industry can help us transcend a sharp split in the literature: improvisational, experiential models of product innovation are appropriate for high-velocity markets while extensive pre-planning and structured processes are appropriate for mature industries (Eisenhardt & Tabrizi, 1995; Kamoche & Cunha, 2001).

Implications for Focal Literatures

Is improvisation valuable for new product development? With the aid of slope analysis I could specify when improvisation is valuable or not for each NPD outcome. Improvisation *increases* cost

efficiency when new product teams rely on internal market information and declarative knowledge and they minimize access to external market information. High levels of both sources and low procedural memory *weaken* the negative impact of improvisation on market effectiveness but they were *not enough* to make it positive. The results contribute to a debate in NPD studies on whether real-time action is a valuable response to unanticipated events (e.g. Cooper & Kleinschmidt, 1986; Eisenhardt & Tabrizi, 1995; Samra et al., 2008). Clearly, the mixed picture of results suggests that this debate should shift from *whether* to *when* and *how* firms can benefit from improvisation.

The difficulty of improvisation to yield market effective new products in a mature setting resonates with cautionary tales about the error-proneness (Davis et al., 2009), inconsistencies (Miner et al., 2001) or biased learning of improvisation (Cunha et al., 1999). The irony of these results is that the structure endowed by procedural memory does not help but, in fact, it hurts the value of improvisation; I explain this issue later in more detail. Future research should consider other factors that might help remove the harmful effects of improvisation in NPD, including transactive memory (Faraj & Sproull, 2004), skill diversity (Kamoche & Cunha, 2001) or team integration processes (Magni et al., 2009). Transactive memory, i.e. team members' awareness of who knows what (Faraj & Sproull, 2000), for example, can help team members link their know-how to create a collective mind (Weick, 1993, 1998) to harness internal, cross-functional expertise, an important driver of market effectiveness.

The role of different sources of market information in improvisation. Moving theory forward, I have offered a set of theoretical ideas about the effects of distinct sources of market information. These two sources have not been studied as a group in the improvisation literature. Hence, I disaggregate the findings of Moorman and Miner (1998b) who examine the moderating role of information flows in general. For example, they found that real-time information does not affect the cost efficiency of improvisation. I, instead, find that internal market information strengthens the positive effect of improvisation on cost efficiency, but external market information cancels this positive effect.

These results also extend a long research tradition on the value of information in innovation. First, the positive moderating role of internal market information for *both* outcomes conflicts with a basic tenet, namely, that internal information trades off between efficiency and effectiveness (Tushman & Katz, 1980; Wong, 2004). Accounting for this disparate result, I suspect that the attention-demanding and the mistake-prone nature of improvisation (Davis et al., 2009) puts a premium on readily available and usable market information. Moreover, the advent of real-time firm information systems for more than decade now (Laudon & Laudon, 1996) has altered this previously alleged trade-off: internal market information can be both timely and easily processed to help teams make cost-efficient and market-effective improvisations.

Second, external market information flows reduce the cost efficiency of improvisation but enhance its long-term market success, which is consistent with research on information use in innovation (Lee & Allen, 1982; Tushman & Katz, 1980). I confirm this finding in a different setting, in which planning and action converge in time. It is also worth noting that the beneficial effects of external sources were captured in the long run but *not* in the short term. One explanation is that these effects of external sources take time to yield results: they boost the novelty in impromptu actions but this novelty typically takes some time to affect market performance. Future research could explore this issue in greater depth.

The role of different types of memory in improvisation. The hypotheses and results unpack the aggregate effects of memory in NPD outcomes (Moorman & Miner, 1998b) and innovative performance of teams (Vera & Crossan, 2005). This study provides more fine-grained results that show it is

declarative memory that helps improvisation boost cost efficiency, while procedural memory makes improvisation detract from market effectiveness.

These results also extend a growing research stream on the value of memory in innovative activities. First, contrary to our hypotheses and prior work (e.g. Cohen & Bacdayan, 1994), procedural memory does not improve the impact of improvisation on cost efficiency. One possible explanation is that the procedural memory used by NPD teams in this sample may have been too rigid to recombine and adapt during improvisation which presents a narrow window of reflection in contrast to pre-planned innovation. It follows that it is useful to refine further the concept of routines with respect to their adaptive quality. Informal procedures and guidelines, for example, may represent more valuable improvisational material than rigid ones (Vera & Crossan, 2005). Future research may explore a richer and more differentiated measure of procedural memory than the rough, general measure used here.

Second, as declarative memory increases the cost efficiency of improvisation, I address a major equivocality in research – prior expertise and knowledge can save time and resources but they can also slow down action, given their abstract nature. Obviously firms' investments in retrieval mechanisms have shifted the balance of value and drawbacks of declarative memory; as firms invest in IT advances (e.g. intranets, blueprints), both access to and the actionability of declarative memory have likely increased (Laudon & Laudon, 1996). Future research could make a contribution by investigating not only types but also *mechanisms* for retrieving declarative knowledge during improvisation. Bartel and Garud (2009) make a contribution in this regard by showing that innovation narratives are an effective retrieval mechanism for translating experience and knowledge to make sense of ambiguous situations and enable real-time action.

Finally, it is puzzling that declarative memory does not matter in the market effectiveness of improvisation. The findings indicate the need for a richer conceptualization distinguishing different *forms* of declarative knowledge. Factual vs. know-why vs. know-who types of declarative memory identified in prior research (e.g. Huber, 1991; Kogut & Zander, 1992) could have a different impact on improvisational types of innovation. For example, know-who types of declarative knowledge and the coordination processes of individual expertise known also as transactive memory (Faraj & Sproull, 2000) could help teams integrate and quickly convert their individual expertise into product specifications during improvisation.

Study Limitations

In interpreting the findings, we should bear in mind that the retrospective subjective perceptions provided by single key informants present some challenges in terms of CMV and accuracy. I discuss here these problems and how I have dealt with them. First, to address CMV, I employed various good practices recommended in the literature (Podsakoff et al., 2003). Different scale formats and anchors (Likert, categorical, semantic differential, and high–low scales) were employed to reduce the respondents' tendency to use their prior responses to answer subsequent questions. For market performance, a key dependent variable, I obtained measures separated temporally from the measures of predictor variables. Moreover, for these longitudinal data, 55% of the respondents were different from the respondents initially providing the predictor variables. To address CMV in the cross-sectional data, I employed Harman's one-factor test and marker variable approach to mitigate the risk of CMV. Though these tests do not rule out CMV completely, they are widely used in the survey literature (Podsakoff et al., 2003).

Second, subjective evaluations of an NPD project after a year could suffer from memory decay, especially when tapping improvisation or use of knowledge resources. Given that the goal,

however, was to obtain *global* measures of memory and information flows, *not* refined ones (e.g. who they talked to, how frequently), memory decay is not a major concern. Likewise, the aim was not to assess improvisation in specific team actions (as was the case in the study of Moorman and Miner, 1998b) or from specific team members; the hypotheses required assessing the extent to which the projects involved deviation from formal plans overall.

Third, subjective evaluations are subject to social desirability given the potentially 'bad' name of improvisation. Though objective measures of performance are more valid than subjective ones, the difficulty of collecting such data at product level explains why this good practice remains rare in NPD research. In alleviating concerns, managerial perceptions have been found to deviate minimally from objective indicators of performance (Dess & Robinson, 1984).

Fourth, multiple respondents can offer a more accurate picture of team activities and processes. Though this is a drawback in this research, single informants have been found to serve as reliable sources, because their estimates fall within 5–10% of one another (Griffin, 1993). These weaknesses could hurt the strength of the findings. On the other hand, the longitudinal data on 132 projects from 132 firms mark progress over previous quantitative studies on improvisation confined to data on action events from two firms (Moorman & Miner, 1998b) or cross-sectional data from a single municipal setting (Vera & Crossan, 2005). Thus, I offer a more generalizable test of improvisation.

Implications for the broader organizational learning research of improvisation

Our literature review has recognized the broader connections between improvisation and organizational learning. We close the loop here reflecting on two key issues – how improvisation affects knowledge creation *and* retention. Being cast as a form of online, localized, short-term learning, generating new knowledge is not an objective of impromptu actions though it can lead to new knowledge with both valuable and harmful outcomes (Miner et al., 2001). While confined to examining NPD outcomes, our theory evokes some explanations that relate to the learning outcomes of improvisation (Cunha et al., 1999) at team level. A key distinction here is the extent to which improvisation promotes exploratory or exploitative learning, which prior work has also discussed in passing (Vera & Crossan, 2005).

The theory can be extended to include examining the impact of improvisation on exploration and exploitation, whose balance and impact on adaptation are important research issues currently (e.g. Gupta, Smith & Shelley, 2006; Kyriakopoulos & Moorman, 2004). For example, the contradictory demands placed on successful improvisation – team members need to deviate from procedural memory but they also need to draw from declarative memory – are somewhat akin to the challenges associated with balancing exploitation and exploration. One conjecture I offer here is that the unifying context of internal sources of market information can increase the likelihood that improvisation promotes exploitative learning, while the diversity in worldviews of external sources can boost exploratory learning. This conjecture can be tested in future research to help us trace the team origins and dynamics of exploitation and exploration.

Relatedly, research can examine how the generated knowledge from local, project-level improvisations can be retained to expand the pool of artifacts, procedures and beliefs contained in organizational memory (Moorman & Miner, 1998a). Brown and Eisenhardt's case studies (1997), for example, have revealed that knowledge from impromptu actions in projects can expand the learning and memory at a firm, strategic level in the form of dynamic but coherent product portfolio choices. They describe two key mechanisms – strong cross-project and cross-functional communication – used to mould these local improvisations into a coherent firm strategy. Miner et al. (2001) have observed

how two firms harvested procedures, artefacts or new understanding resulting from local responses to opportunities or problems. They have observed senior management's efforts to promote ex-post reviews and cross-project learning to retain valuable improvisations in the organizational memory.

These insights from case studies can be the yeast for developing a more precise theory and systematic empirical investigation that help us link local improvisations, meta-routines for harvesting them, and firm learning outcomes across different levels and projects. The reward of such a research strategy will be greater confidence in offering guidelines for developing improvisational competencies in practice.

Conclusion

Improvisation is an increasingly crucial organizational competence for adaptation and innovation. This study shows that, in mature settings, knowledge resources determine when improvisation is valuable in product innovation, extending prior improvisation work confined to fast-paced settings. I find that improvisation *increases* cost efficiency when new product teams rely on internal market information and declarative memory and they minimize external market information, contrary to prior concerns about the efficiency of impromptu action. Though both sources of market information and low procedural memory *weaken* the negative impact of improvisation on market effectiveness, these conditions were not enough to make its impact positive, consistent with traditional concerns.

Appendix

All variables are on 7-point Likert scales unless otherwise noted.

I. Focal study variables

Improvisation (7-point *semantic differential*, Moorman & Miner, 1998b)

- Figured action as we went along—Action followed a strict plan as it was taken
- Improvised in carrying out this action—Strictly followed our plan in carrying out this action
- Ad-libbed action—Not an ad-libbed action

Internal market information flows (Kyriakopoulos & de Ruyter, 2004)

During this project, we:

- regularly received market information, i.e., customer needs, competitive action, from other teams
- systematically received information about market changes from various departments
- systematically collected information on market developments from other concurrent projects
- were kept regularly informed from other sites of our company about market developments
- regularly were updated by customer service people
- regularly updated by salesforce people

External market information flows (Kyriakopoulos & de Ruyter, 2004)

During this project, we:

- systematically discussed with our customers
- regularly consulted our supply chain partners

- systematically contacted our alliances
- regularly discussed with external consultants
- studied systematically successful companies outside our industry*
- consulted regularly outside experts*

Procedural memory (based on Moorman & Miner, 1997): *For this project, my team had:*

- well-defined procedures
- a standard approach
- strong skills*

Declarative memory (based on Moorman & Miner, 1997): *For this project, my team had:*

- a great deal of knowledge
- strong expertise
- knowledgeable people*

New product cost efficiency (new scale, 7 = low, 4 = moderate, 1 = high)

Using the scale, rate this product's costs relative to industry averages for such costs:

- Production costs as a percent of sales
- Promotion costs as a percent of sales
- Total marketing costs as a percent of sales

New product market effectiveness after one year (adapted from Moorman, 1995)

Rate the extent to which the product has achieved the following outcomes during the first twelve months of its life in the marketplace (7 = low, 4 = moderate, 1 = high)

- Market share relative to its major competitor
- Sales relative to its major competitor

New product market effectiveness after two years (adapted from Moorman, 1995)

Rate the extent to which the product has achieved the following outcomes during the first twenty-four months of its life in the marketplace (7 = low, 4 = moderate, 1 = high)

- Market share relative to its major competitor
- Sales relative to its major competitor

II. Control variables

Firm resources (Kyriakopoulos & de Ruyter, 2004)

In comparison with the competition: (7 = high, 4 = moderate, 1 = low)

- Our R&D resources
- Our marketing resources
- Our distribution resources
- Our sales resources

Competitive intensity (Jaworski & Kohli, 1993)

- Competition in our industry is cut-throat
- There are many promotion wars in our industry
- Anything that one competitor can offer, others can match readily
- Price competition is a hallmark of our industry
- One hears of a new competitive move almost every day*
- Our competitors are relatively weak*

Project newness (*categorical variable*, Kyriakopoulos & de Ruyter, 2004)

Indicate the category into which the product best fits:

- Completely new product, new market
- New product line
- Addition to existing line
- Improvement of existing product
- Repositioning of existing product
- Existing product produced at lower cost

* Deleted items

Notes

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- 1 Though some research argues that bricolage (i.e. making do with available resources) is an essential feature of improvisation (Cunha et al., 1999), I follow other research that claims that bricolage happens frequently during improvisation but it can also happen during pre-planning (Miner et al., 2001; Vera and Crossan, 2005).
- 2 Though my review seems to suggest a stark trade-off in efficiency and novelty with internal sources vs external sources, I concede that a black-and-white distinction rarely exists between these two sources. Internal sources, for example, can transmit novel information as members of the same organization might belong to different professional communities and hence develop their own language, worldviews and set of norms. This idea has been widely accepted for more than three decades now as research points that organizational boundaries are permeable as well as complex and dynamic (Sturdy, Clark, Fincham, & Handley, 2009). However, in general, direct contact with external parties, in comparison with internal parties, tends to promote novel thinking (e.g. Moorman, 1995; Von Hippel, 1988; Wong, 2004).
- 3 I use memory and knowledge stocks interchangeably in this paper. This is consistent with a research tradition which stresses organizational memory represents stored knowledge (Moorman & Miner, 1997) contained in organizational storage bins such as structures, ecology and individuals (Walsh & Ungson, 1991). Recent research argues that such 'a functionalist managerialist view' ignores the social and psychological processes of recollecting, and hence memory is broader than stored knowledge (Rowlinson, Booth, Clark, Delahaye, & Procter, 2010: 75). Still, the 'knowledge stock' tradition offers a rich conceptual basis for theorizing how prior knowledge shapes the value of improvisation and it is a part of the broader learning approach to improvisation (Miner et al., 2001; Vera & Crossan, 2005) which I follow.
- 4 The questionnaire was sent in two rounds. The first round yielded 96 responses (ER). To boost the response rate in the second round, I cut down the size of the questionnaire, omitting variables *not* used in this study. The second round yielded 40 responses (LR). Comparing early respondents (ER) to late respondents (LR),

I find no significant differences: improvisation (ER = 3.84, LR = 4.15, $t = 1.31$, NS), procedural memory (ER = 3.89, LR = 3.87, $t = -.11$, NS), declarative memory (ER = 5.03, LR = 5.0, $t = .38$, NS), internal market information (ER = 4.51, LR = 4.38, $t = -.16$, NS), external market information (ER = 4.38, LR = 4.51, $t = .59$, NS), competitive intensity (ER = 4.53, LR = 4.50, $t = .15$), firm resources (ER = 4.27, LR = 4.47, $t = .78$, NS), market performance after one year (ER = 4.69, LR = 4.56, $t = .47$, NS) and new product cost efficiency (ER = 4.10, LR = 3.90, $t = -.90$, NS).

- 5 Comparing non-respondents (NR) to respondents (R) after two years, I find no significant differences on the following measures: improvisation (R = 3.96, NR = 3.98, $t = -.58$, NS), procedural memory (R = 3.92, NR = 3.84, $t = -.29$, NS), declarative memory (R = 5.01, NR = 5.11, $t = .45$, NS), internal market information flows (R = 4.42, NR = 4.60, $t = .91$, NS), external market information flows (R = 4.40, NR = 4.51, $t = .54$, NS), competitive intensity (R = 4.44, NR = 4.61, $t = .91$, NS), firm resources = 4.43, NR = 4.28, $t = .56$, NS), market performance after one year (R = 4.52, NR = 4.77, $t = .86$, NS) and new product cost efficiency (R = 4.07, NR = 3.95, $t = -.57$, NS).

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Author biography

Kyriakos Kyriakopoulos is an Assistant Professor of Marketing and Strategy at ALBA Graduate Business School, Athens, Greece. He holds a PhD from Nyenrode University, the Netherlands. His research focuses on the role of market knowledge, market orientation, and improvisation in firm and interfirm innovation. His work has appeared in *Journal of Marketing*, *International Journal of Research in Marketing*, *Journal of Management Studies*, and *Agribusiness*.

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